CLASS 02 / SEP 1 / U02

CONSTRUCTIVE SOLID GEOMETRY

P2 BEGINS — DESCRIBE A SOLID AS AN EDITABLE SEQUENCE

NOTHING DUE

P2a ISSUED TODAY

WHY NON-DESTRUCTIVE

The same two solids, subtracted. One keeps its history; one throws it away.

DESTRUCTIVE

CUBE − SPHERE → ONE MESH

The sphere is gone. Its radius is gone. To change anything you rebuild from scratch.

NON-DESTRUCTIVE

CUBE + SPHERE → DIFFERENCE → RESULT

Every input is still a live node. Change the radius and everything downstream recomputes itself.

P2a REQUIRES YOUR OPERATION HISTORY TO BE INTACT. YOU WILL BE ASKED TO CHANGE A PARAMETER IN FRONT OF THE CLASS.

TRANSFORMATIONS ARE ARITHMETIC

Three operations you already know, written as what they actually do.

TRANSLATE

= ADD

```
x' = x + dx
y' = y + dy
z' = z + dz
```

```
(2, 1)  +  (3, 0)   →   (5, 1)
```

SCALE

= MULTIPLY

```
x' = x × f
y' = y × f
z' = z × f
```

```
(2, 1)  ×  2        →   (4, 2)
```

ROTATE

= MIX x AND y

```
x' = x·cosθ − y·sinθ
y' = x·sinθ + y·cosθ
```

```
(1, 0)  rot 90°    →   (0, 1)
```

Translation and scale treat each coordinate on its own. Rotation is the only one where x and y have to talk to each other.

THREE BOOLEAN OPERATIONS

Everything in P2a is built from these, applied in order.

UNION

A + B

Everything either solid occupies.

DIFFERENCE

A − B

A, with B carved out of it.

INTERSECTION

A ∩ B

Only where both solids overlap.

ORDER MATTERS. A − B IS NOT B − A, AND YOUR DIAGRAM MUST MAKE THE OPERANDS UNAMBIGUOUS.

P2a asks for all three. In every subtraction, label the solid you keep and the solid you remove. Reverse one once, deliberately, and you will not confuse the operands again.

THE SEQUENCE

What we build together, in order, this class.

01	Start with one primitive; identify node inputs, outputs and parameters.
02	Translate, rotate and scale. Change the order and compare results.
03	Compare union, intersection and difference. Reverse a subtraction and explain what changes.
04	Expose useful inputs and build a short massing sequence.

THE NOTATION

Write the procedure as pseudocode. Formal enough to be unambiguous, plain enough to read aloud.

THE OPERATIONS

```
cube(size)          sphere(r)
cylinder(r, h)

translate(s, x, y, z)
rotate(s, x, y, z)     # degrees
scale(s, factor)

union(a, b)         intersection(a, b)
difference(keep, remove)
```

WORKED EXAMPLE

```
base   = cube(6)
tower  = cube(2)
tower  = scale(tower, 1.5)
tower  = translate(tower, 2, 0, 4)
void   = cylinder(r=1, h=8)

mass   = union(base, tower)
result = difference(mass, void)
```

THREE RULES: EVERY SHAPE STARTS AT THE ORIGIN · EVERY STEP ASSIGNS A NAME · EVERY DIFFERENCE NAMES WHAT IT KEEPS AND WHAT IT REMOVES.

THE SAME THING, WRITTEN OUT

You are not asked to type this. Read it, and check that it says what the nodes you just built say.

```
# nine diagrams, one operation each, in this order

box     = cube(1.0)                  # 1  primitive
ball    = sphere(r=0.65)             # 2  primitive
rod     = cylinder(r=0.30, h=2.0)    # 3  primitive

ball    = scale(ball, 1.15)          # 4  scale
ball    = translate(ball, z=0.15)    # 5  translate
rod_y   = rotate(rod, x=90)          # 6  rotate - degrees
rod_x   = rotate(rod, y=90)          # 6  same operation

solid   = intersection(box, ball)    # 7  intersection
cutters = union(rod, rod_y, rod_x)   # 8  union
result  = difference(solid, cutters) # 9  difference

# 9 keeps solid, removes cutters. draw 9 again at
# building scale: a figure, planting, ground, shadow.
```

WHAT IT MAPS ONTO

parameter

vector

translation

rotation

scale

union

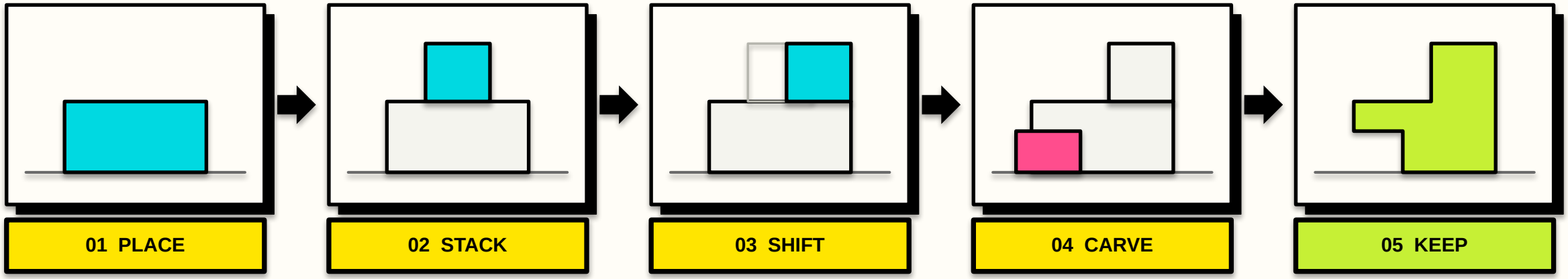
difference

intersection

SAME NOTATION AS THE LAST SLIDE. NINE NUMBERED STEPS, NINE FRAMES — THAT IS YOUR P2a SHEET.

HOW TO DRAW A BIG-STYLE PROCESS DIAGRAM

Explain the building as a row of simple moves: one move per frame, read left to right.



FRAMES — Eight or nine, numbered, all the same size, in one row or a grid.

SAME VIEW — Same camera, same scale in every frame, so only the move changes.

ARROWS — Arrows or numbers carry the order. Nobody should have to guess it.

ONE MOVE — One operation per frame. If a frame needs two verbs, split it.

VERBS — Caption each frame with one verb: place, stack, shift, turn, carve.

KEPT / REMOVED — The new part is the only colour. A removed solid gets its own colour.

Legend above: cyan is the move in this frame, grey is what already exists, pink is the solid being removed, lime is the result.

ISSUED TODAY

P2a — CSG PROCESS AND INSTRUCTIONS

DUE WEEK 4 — Sep 15 / 7%

Explain a CSG object through a BIG-style process diagram with instructions, then 3D print it. This first assignment is the process sheet: an ordered sequence of primitives, transformations and Boolean operations, drawn so another person could rebuild it.

STAGE 1	STAGE 2	STAGE 3
THE SEQUENCE Three primitives. Translate, rotate, scale. Union, difference, intersection.	**THE FRAMES** Eight or nine, numbered. One operation each, read in sequence.	**THE LAST ONE** At building scale: a 1.75 m figure, vegetation, ground, shadow.

REQUIREMENTS — P2a

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01	One 17 × 11 inch sheet laid out as a BIG-style process diagram (in the manner of Bjarke Ingels Group's step-by-step massing diagrams): eight or nine numbered frames, one operation per frame, read in sequence.
02	Use at least three primitives; include translation, rotation and scale, plus union, difference and intersection. Label the kept and removed solids in each subtraction.
03	Draw the final solid at building scale with a 1.75 m figure, vegetation, ground and shadow.
04	Add clear instructions and matching pseudocode beside the diagrams, one step per operation, with the exposed parameters.
05	Refer to Di Mari and Yoo's Operative Design and Di Mari's Conditional Design; name the spatial operation your sequence explores.
06	Submit an editable graph with exposed inputs and all source operations intact.

NOW — BUILD THE SEQUENCE

Rest of the class: a working graph with something you can change in front of me.

- ✓ Object chosen — simple massing, eight or nine clear moves, no complex curves
- ✓ First three primitives placed, each born at the origin and moved from there
- ✓ At least one Boolean operation working, with the kept and removed solids named
- ✓ Two useful inputs exposed, so a variant is one number away
- ✓ Operation history intact — nothing collapsed or applied

LEAVE WITH A GRAPH YOU CAN CHANGE, NOT A MESH YOU CANNOT.

BEFORE NEXT CLASS

NEXT	WEEK 3 — Sep 8 — CSG continued, and 3D printing. P2b, the printed version of your CSG object, is issued.
BUILD	Keep going on P2a, due WEEK 4 — Sep 15: eight or nine frames, the last one at building scale, and instructions with matching pseudocode beside them.
BRING	The editable file with its inputs exposed. Next class we check that the solid is closed and measure it for printing.
READ	Di Mari and Yoo, Operative Design, and Di Mari, Conditional Design. Find the spatial operation your own sequence explores, and name it on the sheet.